- A discrete random variable X is called binomial random variable, if X is the number of successes in n-repeated trials such that

  (i) The n- repeated trials are independent and n is finite

  (ii) Each trial results only two possible outcomes, labelled as 'success' or 'failure'

  (iii) The probability of a success in each trial, denoted as p, remains constant

- The binomial random variable X equals the number of successes with probability p for a success and $q = 1 - p$ for a failure in n-independent trials, has ***a binomial distribution*** denoted by $X \sim B(n, p)$. The probability mass function of X is $f(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, ..., n.$
- If X is a binomial random variable which follows binomial distribution with parameters p and n, the mean μ and variance σ^2 are $\mu = np$ and $\sigma^2 = np(1-p)$.

ICT CORNER

https://ggbm.at/dy9kwgbt or Scan the QR Code

Open the Browser, type the URL Link given (or) Scan the QR code. GeoGebra work book named "**12th Standard Mathematics Vol-2**" will open. In the left side of work book there are chapters related to your text book. Click on the chapter named "**Probability Distributions**". You can see several work sheets related to the chapter. Go through all the work sheets.

12th_Maths_Vol 2_EM_CH 11_Probability Distributions_29-05-20.indd 223 16-12-2022 16:52:54